

Evis Plaku

PhD Candidate · Artificial Intelligence & Intelligent Systems

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PROFILE SUMMARY

My work spans autonomous robotics, machine learning, and intelligent systems; as a researcher, an educator, and a practitioner who values rigour and real-world applicability in equal measure. Across a decade of research, teaching, and building, I have worked to bridge formal AI methods and real autonomous behaviour, driven by a long-term interest in systems that require ever less human guidance.

EDUCATION

PhD Candidate in Artificial Intelligence & Intelligent Systems <i>Metropolitan University of Tirana, Albania</i> Dissertation: Enhancing Autonomy of Mobile Robots Operating in Complex Environments	Oct 2022 – present
Advanced Studies in Machine Learning & Robotics <i>Catholic University of America, Washington D.C., USA</i> New Millennium Research Assistantship recipient	Jan 2016 – Sep 2016
M.Sc. in Artificial Intelligence and Robotics <i>University of Freiburg, Germany</i> Thesis: <i>Inverse Reinforcement Learning for Mobile Robot Navigation</i>	Oct 2010 – Jul 2013
M.Sc. in Computer Science <i>University of Tirana, Albania</i> Thesis: <i>Game Playing through Machine Learning and High-Dimensional Search</i>	Oct 2008 – Jun 2010
B.Sc. in Computer Science <i>University of Tirana, Albania</i>	Oct 2005 – Jun 2008

PUBLICATIONS

Refereed International Journals

- [J1] E. Plaku, A. Çela, E. Plaku. “Generating Diverse Trajectories Using Roadmap Search and Sampling-Based Motion Planning.” *IEEE Access*, May 2026. doi:10.1109/ACCESS.2026.11506360
- [J2] E. Plaku. “Semantic-Aware Graphs for Safe Path Planning in Urban Driving Environments.” *International Journal of Control, Automation and Systems*, 2026. doi:10.1007/s12555-026-00116-4
- [J3] E. Plaku, A. Çela, E. Plaku. “Enhancing Robot Navigation: Integrating Safety Zones into Path Planning.” *Lecture Notes in Electrical Engineering, Springer*, 2024.
- [J4] E. Plaku, E. Plaku, P. Simari. “Clearance-Driven Motion Planning for Mobile Agents with Differential Constraints.” *Robotica*, Cambridge University Press, 2018.
- [J5] E. Plaku, E. Plaku, P. Simari. “Direct Path Superfacets: An Intermediate Representation for Motion Planning.” *IEEE Robotics and Automation Letters*, 2017.

Refereed International Conferences & Proceedings

- [C1] E. Plaku, G. Lici, S.-I. Niculescu, A. Çela. “Safe Urban Route Planning via Semantic Graphs and Neural Cost Shaping.” *34th Mediterranean Conference on Control and Automation (MED)*, 2026.
- [C2] G. Lici, E. Plaku, S.-I. Niculescu, A. Çela. “An Adaptive-Horizon Model Predictive Control Approach for Robot Navigation in Dynamic Environments.” *34th Mediterranean Conference on Control and Automation (MED)*, 2026.
- [C3] E. Plaku. “Motion Planning with Large Language Models: Language-to-Paths Guidance over Superfacets.” *3rd International Conference on Information Technologies and*

Educational Engineering (ICITEE), 2025.

- [C4] E. Plaku, A. Çela, N. Civici. “**Safety Analysis for Autonomous Mobile Robot Navigation Using Traffic Sign Recognition, Dynamic Object Detection, and Semantic Segmentation.**” *International Conference on Computing, Networking, Telecommunications and Smart Applications (CoNTESA)*, pp. 5–10, Dec. 2024. doi:10.1109/CoNTESA64738.2024.10891271
- [C5] E. Plaku, K. Jahaj, A. Çela, N. Civici. “**A Machine Learning Framework for Automated Albanian News Article Title Classification.**” *18th International Conference on Innovations in Intelligent Systems and Applications (INISTA)*, 2024.
- [C6] E. Plaku, A. Çela, E. Plaku. “**Robot Path Planning with Safety Zones.**” *20th International Conference on Informatics in Control, Automation and Robotics (ICINCO)*, 2023.
- [C7] E. Plaku. “**Inverse Reinforcement Learning for Computing Navigation Actions From Demonstrations.**” *International Conference on Computing, Information Technology and Business Applications*, 2017.

ACADEMIC & PROFESSIONAL EXPERIENCE

Lecturer and Researcher

Oct 2022 – present

Metropolitan University of Tirana, Albania

Teaching AI and software engineering; supervising student theses; conducting research on autonomous motion planning at the intersection of AI, machine learning, and classical planning algorithms.

Machine Learning Instructor

Feb 2022 – present

Holberton School

Hands-on instruction in applied machine learning, deep learning, and AI tools for industry-oriented learners.

Freelance Academic Author & Video Lecturer

IU International University of Applied Sciences, Germany

Authored three full coursebooks; produced video lectures for over ten university-level courses in computer science and artificial intelligence.

Lecturer in Computer Science & AI

Oct 2018 – Sep 2022

Canadian Institute of Technology, Albania

Visiting Lecturer

Mar 2021 – Sep 2021

Polytechnic University of Tirana, Albania

Algorithms and Advanced Programming (Labs).

Lecturer in Computer Science

Mar 2017 – Sep 2018

Metropolitan University of Tirana, Albania

COURSES TAUGHT

- **Metropolitan University of Tirana:** Object-Oriented Programming; Data Structures and Algorithms; Deep Learning; Advanced Programming; Computer System Organization; Management Information Systems; Operating Systems; Planning & Managing Sustainable Projects.
- **Canadian Institute of Technology:** Fundamentals of Programming I & II; Computer Applications.
- **Polytechnic University of Tirana:** Algorithms and Advanced Programming (Labs).
- **University of Freiburg, Germany (Teaching Assistant):** Foundations of Artificial Intelligence; Theory of Algorithms; Software Design, Modeling & Analysis in UML; Introduction to Mobile Robotics; Theoretical Computer Science II.

TECHNICAL SKILLS

Robotics & Motion Planning	ROS/ROS2, Gazebo, MoveIt, SLAM, kinodynamic planning, sampling-based planning (RRT, PRM), safety-constrained planning, workspace decomposition, graph search, MPC
Machine Learning & DL	TensorFlow, Scikit-learn, CNNs, transfer learning, object detection & segmentation, semantic segmentation, reinforcement learning, imitation learning
AI Systems & LLMs	LLMs, prompt engineering, RAG systems, HuggingFace Transformers, LangChain, embeddings, OpenCV, computer vision pipelines
Scientific Computing	NumPy, SciPy, Pandas, Matplotlib, Seaborn, MATLAB
Programming & Engineering	Python, C/C++, Java Git, Docker, REST APIs, FastAPI

LANGUAGES

Albanian	Native	
English	C2 — Proficient	TOEFL & SAT certified; full academic and professional fluency
Italian	C1 — Advanced	CELI 3 Certification
German	B1 — Intermediate	Goethe Institute Certification